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Short communication

DeepFusion: A simple way to improve traditional multi-view stereo methods using deep learning

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ABSTRACT

In this paper, we propose a simple deep learning based method, "DeepFusion", to improve the results from traditional multi-view stereo (MVS) methods. In our method, we first design a deep convolutional neural network to predict the confidences of the depths produced by the traditional MVS method, which takes the colors, depths, normals and costs as inputs. In particular, we convert the depths into the normalized inverse depths to enhance the accuracy and generality of the network. In addition, we use the plane sweeps in the disparity space to get the top-k matching costs in order to determine the distinctiveness among all costs. Then we propose a novel approach to accurately fuse all depth maps into the final point cloud, which balances the geometric consistency and the predicted confidences. Experimental results on several well-known datasets demonstrate the effectiveness and the generalization ability of the proposed method.

1. Introduction

Multi-View Stereo (MVS) methods generally consist of five steps, namely, structure from motion (SFM), dense reconstruction, mesh reconstruction, mesh refinement and mesh texturing. The SFM takes in a set of images and recovers the intrinsic and extrinsic parameters of the cameras used. Next these parameters are fed to the process of dense reconstruction to estimate the depth maps for each of the images, which are then fused into a dense point-cloud. The mesh reconstruction step takes those points to form a surface mesh and the mesh refinement tries to optimize this mesh by reducing the re-projection errors. The mesh texturing gives colors to the mesh to improve the visual experience. Among these steps, the accuracy of the dense reconstruction has a large impact on the final MVS result since wrong points will result in erroneous faces and it is often hard for the refinement step to pull them back to the right positions.

Although there exist some nice mesh reconstruction algorithms [1–3] that can handle outliers quite well, they are still not stable enough when dealing with untextured regions or edges of objects. Thus, existing MVS methods adopt different means to filter these outliers in the dense reconstruction. Most of traditional MVS methods [4–6] rely on the geometric consistency and the camera-point visibility intersections to identify outliers. Some other work encourages the depth smoothness using a median filter [7] or directly removing those disconnected segments [5,8]. However, all these strategies may bring bad effects to the point-cloud, for instance, encouraging geometric consistency could cause the decrease of completeness especially when the set of images have a relatively low rate of overlap with each other.

Recent researches on deep convolutional neural networks (CNNs) have demonstrated that using CNNs to predict depth maps can dramatically improve the performance of MVS since CNNs have mush larger receptive fields which decrease the probability of getting wrong estimates [9–11]. On the other hand, these CNN-based MVS methods all suffer from the large memory requirement due to building the cost volume, which leads to limitation in application scenarios with high-resolution images. We note that CNNs also have been popularly used to predict the confidence maps without the need of building the cost volume, and these confidence maps could be further used in the traditional MVS methods to improve their performance. Such combination may not only get rid of the memory restriction brought by the cost volume but also maintain the advantage of large receptive field at the same time. The confidence prediction has been widely used in binocular stereo matching [12–15], but unfortunately, only very few researches have done on MVS [16–18], and there still lacks a specially designed architecture with good accuracy and generality for MVS confidence prediction.

In this paper, we develop a deep learning based method, called "DeepFusion", to improve the results from traditional MVS algorithms. Our main contributions can be summarized as two

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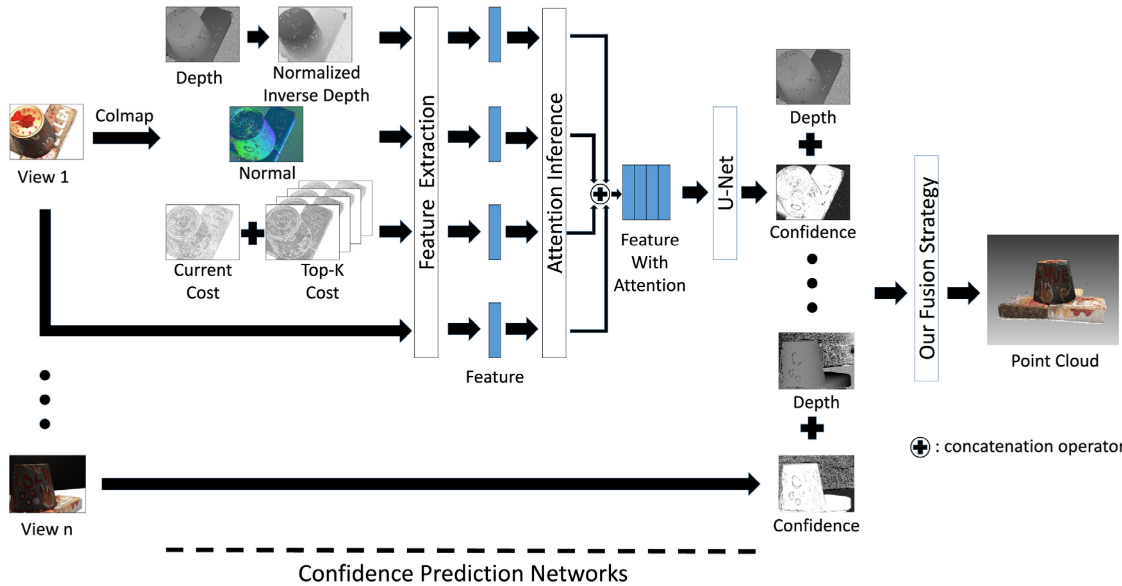


Fig. 1. Overview of the pipeline of "DeepFusion". We first use Colmap to generate the depth map, the normal map and the cost map for each view. In particular, we convert depths into normalized inverse depths and locally sweep around the current depths to get top-k costs. All of these plus colors are then concatenated together using the attention module to form a feature map with attention, which is fed to an U-Net to produce a confidence map. In the end, we design a novel approach to fuse all depth maps into the final point cloud, which balances the geometric consistency and the confidence prediction.

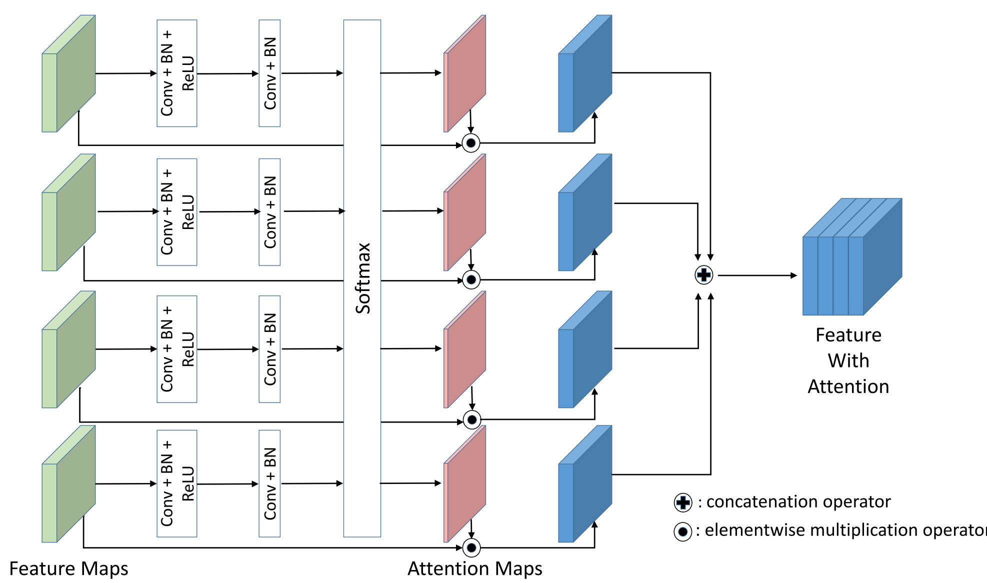


Fig. 2. The attention inference model. The four feature maps are fed to the network to learn the one-channel attention maps which are then embedded into the feature maps using an elementwise multiplication operator. The four new feature maps are concatenated to form the feature map with attention.

aspects: (1) we design a deep convolutional neural network for confidence prediction based on the outputs of a traditional MVS method, which can deal with various scenarios and datasets with good generality; (2) we propose a simple but effective approach to utilize the predicted confidence maps and the geometric consistency to accurately fuse all depth maps into the final pint cloud. We choose the PatchMatch-based method, Colmap [6], to be the traditional MVS algorithm tested in our method since it works on various datasets and has an open-source code to reproduce. Our method is trained and tested on the DTU dataset [19] and the results demonstrate the great improvement it brings. We also test our method on the Tanks and Temples dataset [20], the Colmap datasets [6] and an arial image set to show the good generalization ability of our method.

2. Related work

The depth estimates obtained from traditional MVS methods are usually noisy since most of them only look for local optimization using the photometric consistency constraint. Existing methods for improving the quality of dense reconstruction in traditional MVS methods can be basically divided into two categories as reviewed in the following.

2.1. Conventional approaches

To filter out these noises, most conventional methods still choose to encourage the geometric consistency in certain ways. PMVS [4] divides an image into cells and enforces the reconstructed patches in the same cell to be consistent with each other.

(a) Cross entropy loss   (b) Focal loss   (c) Ground-truth

Fig. 3. The results on a model from the DTU validation dataset show the difference between using the basic cross entropy loss and using the focal loss. The gray color corresponds to the low confidence and the dark color means there is no depth offered. The focal loss performs much better on details.

OpenMVS [5] filters each of the depth maps using its neighboring depth maps. Specifically, the neighboring depth maps are first wrapped to the reference view. The inconsistency with neighboring depths will decrease the confidence of the reference depth while the consistency will increase the confidence. Once the reference depth is considered to be reliable, the neighboring depths which have conflict with the reference depth will be set to wrong. Colmap [6] embeds the geometric consistency into its process of depth maps estimation. It repeats the depth maps reconstruction twice, and when it propagates the depths at the second time, it does not propagate those depths that are inconsistent with the view it selects. ACMM [7] applies a median filter of size 5 $\times$ 5 to the final depth maps. TAPA-MVS [8] adopts a classical speckles filter to remove the discontinuous depths and uses the weighted median of neighboring depths to fill the missing parts. However, all these strategies are hand-crafted and lacks the generality, and further could not deal with noises at the edges of the reconstructed objects. In addition to these local refinement, global optimization has also been used to improve the accuracy. DeepMVS [21] converts the MVS problem into multi-class labeling and applies the full-connected conditional random field [22] to improve the quality of the raw depth maps. By adding the regularization term to the optimization function to ensure smoothness, outliers are expected to be filtered out [11,23]. This kind of approaches can help to reject outliers but often need several iteration and are time consuming.

2.2. CNN-based approaches

CNN-based methods have been recently developed for improving traditional MVS methods. The confidences predicted by CNNs can help improve the disparity maps in binocular stereo matching as shown in [12–15,24–26]. Early methods [12,13] cut images into patches of regular sizes and feed these patches into the networks to predict the confidences at the center position of the patches. The confidences are then incorporated into the Semi-Global Matching (SGM) to improve the initial disparity maps [12]. However, the optimal size of the patch can differ significantly among different datasets and it is not robust to rely on only local features. To extract global features, several recent methods [14,15] directly feed the whole images and disparity maps to networks. Moreover, to make better use of the tri-model (colors, disparities and matching costs), LAF-Net proposes [15] attention inference networks to concatenate these three inputs together. Compared to the prosperous development of confidence prediction in binocular stereo matching, there only are very few studies related to its application on MVS [16–18]. Kuhn et al. [16] use color, normal and the number of successfully matched images of each pixel to predict the corresponding confidence. The confidences are then used to guide two processes: filtering and refinement on the depth maps. The depths are not taken into account since they are not normalized and instead the normal maps are used in this method. However, the normals estimated by PatchMatch-based methods are usually less accurate compared to the depths since they have one more degree of freedom than depths, which impact the result of confidence prediction. Meanwhile, the number of successfully matched images can reflect the geometric consistency but has little generality since different datasets have different levels of overlaps. MG-MVS [17] directly deals with the surface mesh and its goal is to remove erroneous surface which will mislead the following process in its pyramid architecture. CLD-MVS [18] trains a network that takes depth patches as input and outputs the confidence of the center depth. It then applies a confidence-driven and boundary-aware interpolation to improve the completeness of depth maps. As mentioned above, the drawback of these patch-based methods is that they do not consider global information. Thus, it is in urgent need to design an architecture with good generality and accuracy for the MVS confidence prediction.

3. Proposed method

In this section, we describe our method "DeepFusion", see Fig. 1 for an illustration of its pipeline. We first introduce our confidence prediction network for MVS and the motivation of designing such an architecture, then we present an effective way on how to integrate the predicted confidences into the fusion step of forming the final point cloud.

3.1. The confidence prediction network

In consideration of the great success of confidence prediction networks in binocular stereo matching in recent years, we imitate some well-performed networks and transplant effective models from them to design our confidence prediction network that fits the needs of MVS.

Inputs of the network. A tri-model was proposed in LAF-Net [15] to extract features from colors, disparities and top-k costs, and has demonstrated its effectiveness through various experiments. Colors are directly from the given set of images. Different from binocular stereo matching, we only can obtain estimated depths directly from traditional MVS methods. Unnormalized depth values lack generality and are not suitable to be directly fed into the network as showed in DeepC [16]. In addition, estimated depths from MVS are often much more accurate compared to other outputs and could effect the performance of the confidence prediction network more significantly, which will be demonstrated later in our experimental section. Before normalizing the depth values, let us first pay close attention to the difference between the disparities and the depths. The changes of disparities from closer objects are larger than those from farther objects, which

Table 1
Ablation study for our network with different combination of inputs on the DTU validation set using AUC measurement. The optimal value corresponds to the AUC of the ground-truth. The smaller the score is, the better the performance is.

Model	Normalized inverse depth	Matching cost	Focal loss	AUC
A		✓	✓	0.07401
B	✓		✓	0.03939
C	✓	✓		0.03904
Full	✓	✓	✓	**0.03839**
Optimal				0.02150

will make the network pay more attention to closer objects. This character should also exist in the confidence prediction network since depth estimation of closer objects is much more precise. Therefore, we better convert the depths produced from traditional MVS methods (such as Colmap) into inverse depths for feature extraction in the network. Moreover, compared with the disparities which always fall within the image coordinate system, the depth values are actually in the camera coordinate system. Although disparities lack the scale information of the scene compared to depths, feeding disparities to networks still can yield good results for binocular stereo matching, which means that the scale information in fact only has little effect on performance of the confidence prediction network. Thus we can use the depth range to normalize the depths to eliminate the effects brought by scale difference between datasets. Based on these arguments, we transform the depth d to the normalized inverse depth d_{ni} using the depth range $[d_{min}, d_{max}]$ of the current view as follows:

$$d_{ni} = \frac{\frac{1}{d} - \frac{1}{d_{max}}}{\frac{1}{d_{min}} - \frac{1}{d_{max}}}. \tag{1}$$

To eliminate the influence caused by outliers, we adopt the depth range computed by Colmap [6] and set those depths out of this range to infinity.

Besides the depths, the top-k matching costs are also important to confidence prediction since they can tell the network whether a depth value is reliable enough. However, there is no cost volume in PatchMatch-based methods which is used to get the top-k costs in LAF-Net. MG-MVS [17] perturbs the current depths for k times to get the top k costs. It is a compromise since random perturbation also produces uncertainty. To get the top-k costs in PatchMatch-based MVS, we adopt a local plane sweeping strategy similar to [27]. We sweep each depth using the corresponding normal produced by Colmap. To deal with the scale difference in depth, we compute an average baseline following [28] and convert those depths into disparities. For each pixel p in the reference image r, we compute the baseline $b_{r,t}$ between r and the target views V_r^p with top-five probability predicted by Colmap and average them to get b_r^p:

$$b_r^p = \frac{1}{|V_r^p|} \sum_{t \in V_r^p} b_{r,t}. \tag{2}$$

To maintain generality, we mark a view as invalid if its probability is under 0.1 and filter out such views from V_r^p. If there is no view left after filtering, we set all the top k costs to 2 (the cost ranges from 0 to 2 and the smaller, the better). We compute the matching costs with different disparities around the current estimate and pick up the top-k costs from them. The top-k costs together with the current cost are all feed to the confidence prediction network.

Similar to DeepC [16], we also feed the normals to the network, which can give a constraint to the neighboring depths since it ought to correspond to the gradient of depths.

Architecture of the network: The four inputs are separately fed to a feature extraction model which consists of three sub-blocks (2D convolution, Batch Normalization and ReLU) to get four feature maps with 32 channels. Different from DeepC which directly concatenates the feature maps, we leverage the attention inference model [15] which is widely used in other fields [29,30] to concatenate the four types of inputs. As shown in Fig. 2, the attention inference model actually infers the weight of each kind of feature, which allows the network to pay more attention to the realiable features. We then adopt an U-Net architecture [31] following the work of ConfNet [14] and DeepC [16] since U-Net considers both local and global features and is able to output a confidence for each pixel. The feature map with attention is fed to the U-Net which has four poolings and four fractionally-strided convolutions. Each of them is followed by two sub-blocks with 2D convolution, Batch Normalization and ReLU. See Fig. 1 for an illustration of the architecture of our confidence prediction network.

Loss function: The PatchMatch-based methods have developed for decades and can perform well in most cases. The majority of depths estimated by Colmap are correct, which may cause the extreme imbalance in the training data. Moreover, most depths are easy to be labeled as right or wrong, which could make the network neglect those that are hard to be judged. To deal with this problem, we utilize the idea of focal loss [32] which is proposed for solving the class imbalance problem. Another issue we may face is the incompletion of ground-truth. We could not tell whether a depth estimated by Colmap is right or wrong when the dataset does not offer the ground-truth. Although most of these kind of depths are wrong since there indeed exists no object, some depths still could be right. The corresponding points with respect to them are missing in the point-cloud provided by the ground-truth. Thus we give the area of these points a lower weight (0.5 in our training) so that the network can still filter out outliers while does not delete those depths that should exist. The loss function is then defined as:

$$F(p) = -w(1-p_t)^{\gamma} log(p_t) \tag{3}$$

with

$$p_t = \begin{cases} p, & if\ y = 1, \\ 1-p, & otherwise, \end{cases} \tag{4}$$

where w is the weight of each pixel, y is the ground-truth label while p is the confidence predicted by the network, γ corresponds to the degree of focal to those hard samples and we set it to 2.

3.2. Fusion strategy for depth maps

The quality of the point-cloud has a strong impact on the results of reconstruction and some other down-stream point cloud tasks [33,34]. DeepC-MVS [16] and CLD-MVS [18] all leverage the confidences to interpolate the missing depths for each view in order to improve the completeness of the point-cloud. Their interpolation is based on the assumption that the missing depths are always from planar objects. As mentioned before, the dense reconstruction is only a step in the MVS pipeline. To judge whether a point-cloud is a good input for mesh reconstruction, we should focus on whether it can help to improve the completeness and accuracy of the mesh. Actually, the planar structure can be easily reconstructed by the mesh reconstruction method [35] which is widely used today if the contours exist in the point-cloud

(a) Image (b) Confidence filtering (c) Our fusion strategy

Fig. 4. The effects of our fusion strategy on an model from the aerial image dataset. Using the confidence filtering strategy can improve accuracy but it may also delete details. Unlike the planar structure, these details cannot be properly estimated by mesh reconstruction. Our fusion strategy can help preserve these details while still maintaining good accuracy.

and there is little outliers around the plane. Whether there exist correct estimates on the plane often has little influence on the final MVS result. That is to say, to improve the completeness of the mesh, we need to obtain complete contours and to filter out the outliers between those contours. In addition to effect on the planar objects reconstruction, outliers can also produce erroneous surfaces or cause inaccurate edges, which is hard for mesh refinement to calculate the correct local optimums. Here, we propose a new fusion method which makes use of confidences to delete outliers. DeepC-MVS$_{fast}$ directly filters those depths with low confidence from depth maps. However, when images do not overlap each other much, this strategy will lead to great decrease in completeness since a single wrongly predicted confidence can cause the fused point loses enough views to support it. Thus, we add the confidences to the fusion step, and the confidence of a fused point is the median of confidences of its supporting depths. To respect both the geometric consistency and the predicted confidence, we increase the confidence of a point with high geometric consistency. Then, the confidence c_p of a fused point is defined as:

$$c_p = \omega^{(|V|-V_{min})} Median(c_v), \quad v \in V, \tag{5}$$

where V contains all the supporting views and V_{min} is the minimum number of supporting views, ω corresponds to the weight of geometric consistency and we set it to 1.1 in all our experiments. We then filter out those points with a confidence lower than 0.5.

4. Experiments

Datasets. The datasets used in our experiments include the DTU dataset [19], the Tanks and Temples benchmark [20], the Colmap dataset [6], and a self-collected aerial image dataset.

Training. We train the confidence prediction network of our method "DeepFusion" on the DTU dataset only. Following SurfaceNet [36], we divide the dataset into three parts (training, validation and evaluation). For the sake of efficiency, we downsample the images and cut them to the size of 640 × 512 while training. We first run Colmap and generate inputs for our method. We set the geometric consistency option in Colmap to be false since it sometimes cause erroneous deletion and we want to leave the filter task to our method. To get the ground-truth for training, we compute the uncertainties for each depth at each pixel p in the image r using the model from [37]:

$$\sigma_P = \sigma_p \frac{d_p^2}{f_r b_r^p} \sqrt{2}, \tag{6}$$

where d_p is the depth at p, f_r is the focal length of r, b_r^p is the average baseline obtained using Eq. (2), and σ_p is the pixel uncertainty which we set to 1 in all experiments. If the difference between the estimated depth and the ground-truth depth is bigger than σ_P, then we deem the confidence of this depth is 0. The confidence prediction network is trained for 20 epochs using ADAM [38] with a batch-size of 2. The learning rate is set to be $1.0e-3$ initially and we multiply the current learning rate with 0.9 after each epoch.

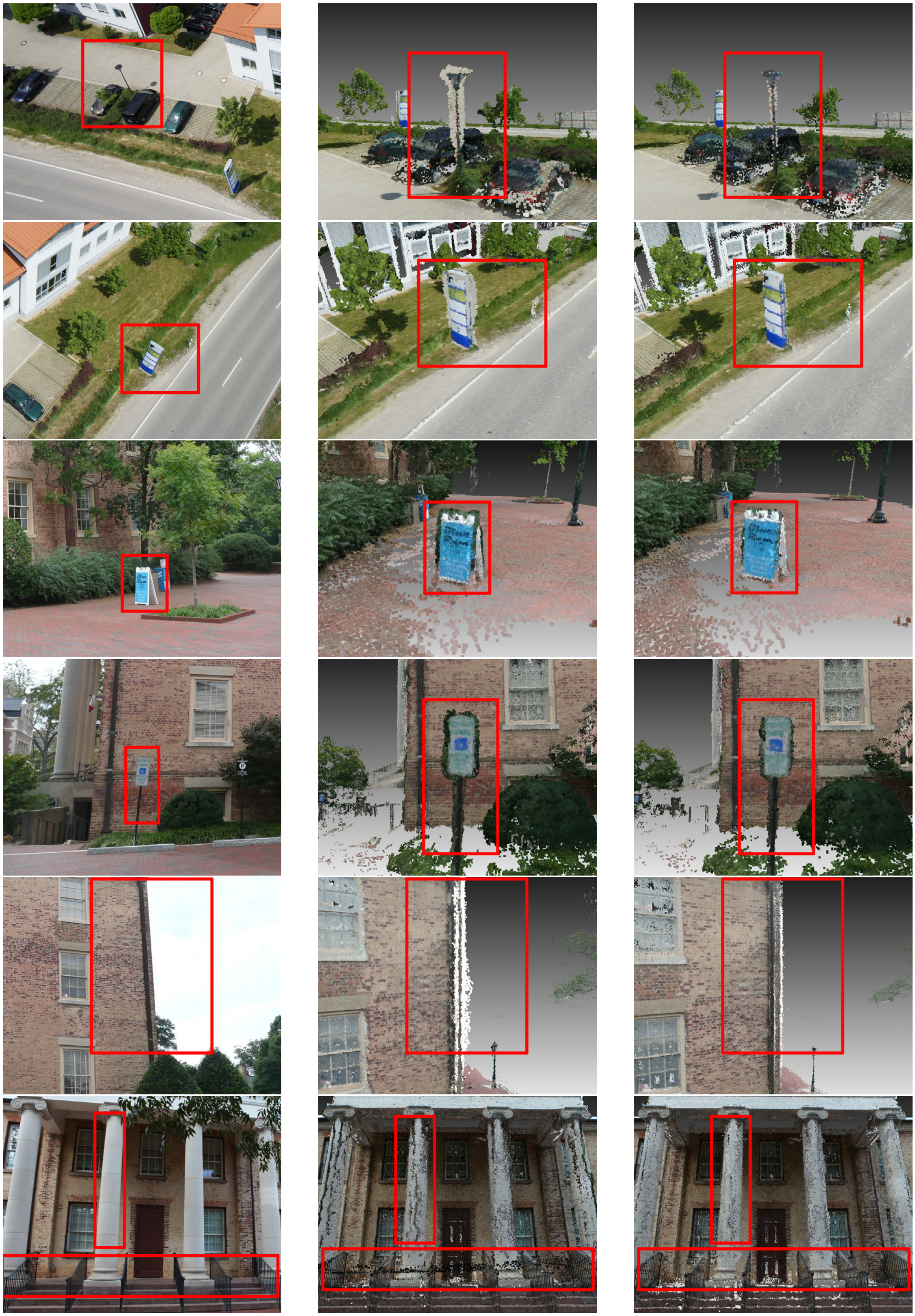
4.1. Ablation study

In this section, we perform an ablation study to show the effects of confidence prediction network, loss function and fusion strategy in our method. Following [39], we compute the area under the curve (AUC) to measure the accuracy of confidence prediction. Specifically, we sort the confidences in the decreasing order and select the top 5% to compute the error rate. We repeat the same computation for top 10% and so on to get the receiver operating characteristic (ROC) curves. The AUC corresponds to the area formed by the ROC curves. We then fuse the depth maps to a point cloud and following R-MVSNet [11] we compute mean accuracy, mean completeness and their average (overall score) of the point cloud as measurements for evaluating the fusion strategy and the whole pipeline of DeepFusion.

Effects of the inputs. Table 1 reports the performance comparison of several model variants of our method with different combination of inputs. DeepC-MVS uses normal maps as input to displace the depth maps since depth maps are not normalized. However, compared to normals, depths estimated by PatchMatch-based methods are always more precise. We compute the normalize reversed depths and feed them to the confidence prediction network and find that this way can help to improve the confidence prediction very much. The counter maps proposed by DeepC-MVS correspond to the reliability of depth estimates, but they rely heavily on the overlapping rate of input images and could differ much between datasets, which may cause the loss of generality. To take the reliability into consideration, we compute the top-k costs and feed them to the network together with the current cost, which also can help to decrease the AUC while do not affect the generality.

Effects of the loss function. Applying the focal loss can help preserve more details since the loss function will give them more weight. Fig. 3 shows the improvement on details by using the focal loss over the basic cross entropy loss on the DTU validation dataset. There is only a very small decrease of AUC brought by the focal loss since the details only make a tiny portion of the validation data.

Effects of the fusion strategy. Wrong confidence prediction of depth is always unavoidable and directly filtering the depth maps using confidence may lead to insufficient support for some points, which then cause wrong deletion. Instead of completely relying on confidence, our fusion strategy also respects the geometric consistency. As demonstrated in Table 2, our fusion strategy can maintain completeness with only a little decrease of accuracy,



(a) Image (b) Colmap (c) DeepFusion (Ours)

Fig. 5. Visual comparisons of performance of our method with Colmap. The top two rows are from the aerial dataset and the rest of rows are from the Colmap dataset.

Table 2
Ablation study for our fusion strategy on the DTU evaluation dataset. The smaller the score is, the better the performance is.

Method	Mean accuracy	Mean completeness	Overall
Confidence filtering	**0.334**	0.548	0.441
DeepFusion w/o geom	0.345	0.521	0.433
DeepFusion	0.357	**0.502**	**0.429**

Table 3
Quantitative comparisons of performance of different MVS methods on the DTU evaluation dataset. The smaller the score is, the better the performance is.

Method	Mean accuracy	Mean completeness	Overall
Gipuma [40]	**0.274**	1.193	0.734
tola [41]	*0.343*	1.190	0.767
furu [4]	0.612	0.939	0.776
camp [42]	0.836	0.555	0.696
Surface-Net [36]	0.450	1.043	0.746
MVSNet [43]	0.396	0.527	0.462
R-MVSNet [11]	0.385	**0.459**	**0.422**
Colmap [6]	0.377	0.538	0.457
DeepFusion (Ours)	0.357	*0.502*	*0.429*

Table 4
Quantitative comparisons on generalization ability of different methods on the Tanks and Temples dataset. The higher the score is, the better the performance is.

Method	Mean precision	Mean recall	Mean F-score
OpenMVS [5]	41.16	*67.26*	50.47
PCF-MVS [28]	*50.04*	58.85	*53.39*
Surface-Net+ [44]	**51.86**	50.30	49.38
Point-MVSNet [45]	41.27	60.13	48.27
Dense R-MVSNet [11]	39.80	**71.96**	50.55
R-MVSNet [11]	43.74	57.60	48.40
Colmap [6]	43.16	44.48	42.14
DeepFusion (Ours)	46.72	67.16	**54.02**

which result in an improvement of overall performance. Notice that the completeness difference is not caused by planar areas which are easy to be recovered by mesh reconstruction, the wrongly deleted points are more likely to be details or at non-planar regions as shown in Fig. 4.

4.2. Comparison with existing methods

On the DTU benchmark. We compare our method with the original Colmap on the DTU evaluation dataset, as well as some other existing methods including traditional and learning-based MVS methods, to demonstrate the performance of our method. The quantitative comparison results are reported in Table 3. First, it is easy to say that our method improves both accuracy and completeness of the original Colmap. This is partly because the geometric consistency check in Colmap does cause wrong deletion. Our method preserves the right points as many as possible while still maintains good accuracy. We also would like to remark that it is quite straightforward to transplant our method to other traditional baseline algorithms. Second, our method achieves higher accuracy (ranked the third) than most of these comparison methods and the state-of-the-art performance in the terms of the completeness and the overall score (both ranked the second).

Generalization ability. To demonstrate the generalization ability, we first test our method (still with the network trained on the DTU dataset) on the Tanks and Temples benchmark. This dataset contains two sets: the intermediate set and the advanced set. Different from the DTU dataset, the Tanks and Temples dataset only provides images and one needs to run the SFM to obtain the parameters of cameras. We only test our method on the intermediate set since Colmap often fails to produce the right camera poses for many samples of the advanced set. The quantitative comparison results are reported in Table 4, where the measurements of mean precision, mean recall and mean F-score are used. It is observed that our method significantly improve the results of Colmap in both precision and recall. Among all traditional and learning-based methods for comparison, our method achieves remarkable performance in terms of precision and recall, which result in the top-ranked F-score. We also test our method on the Colmap dataset and a self-collected aerial image dataset and compare the point clouds with those from the original Colmap. Fig. 5 presents visual comparisons of some models reconstructed by our method and the original Colmap. As we can see, our method contains much less noises while still preserving the details well.

5. Conclusion

In this paper, we first propose a specially designed network architecture for confidence prediction of MVS. In particular, we convert the depth values into normalized inverse depths in the inputs to improve generality, and the current depths are locally swept in the disparity space to get the top-k costs to reflect the distinctiveness of the current costs. We adopt the attention module to concatenate all input data, namely depths, costs, normals and colors, and then feed them to a simple U-Net to get the pixel-wise confidence prediction. In order to alleviate the influence of wrong predictions, both the predicted confidences and the geometric consistency are taken into account to develop a novel strategy for fusing all depth maps into the final point cloud. The experimental results on several well-known datasets demonstrate that our method can achieve state-of-the-art accuracy and completeness at the same time and possesses excellent generalization ability.

CRediT authorship contribution statement

Yuesong Wang: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing. **Keyang Luo:** Data curation, Writing - original draft. **Zhuo Chen:** Software, Writing - original draft. **Lili Ju:** Methodology, Writing - original draft, Writing - review & editing. **Tao Guan:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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